A multicenter randomized controlled trial of a plant-based nutrition program to reduce body weight and cardiovascular risk in the corporate setting: the GEICO study

Background/objectives: To determine the effects of a low-fat plant-based diet program on anthropometric and biochemical measures in a multicenter corporate setting. Subjects/methods: Employees from 10 sites of a major US company with body mass index ■25 kg/m2 and/or previous diagnosis of type 2 diabetes were randomized to either follow a low-fat vegan diet, with weekly group support and work cafeteria options available, or make no diet changes for 18 weeks. Dietary intake, body weight, plasma lipid concentrations, blood pressure and glycated hemoglobin (HbA1C) were determined at baseline and 18 weeks. Results: Mean body weight fell 2.9 kg and 0.06 kg in the intervention and control groups, respectively ($P<0.001$). Total and low-density lipoprotein (LDL) cholesterol fell 8.0 and 8.1 mg/dl in the intervention group and 0.01 and 0.9 mg/dl in the control group ($P<0.01$). HbA1C fell 0.6 percentage point and 0.08 percentage point in the intervention and control group, respectively ($P<0.01$). Among study completers, mean changes in body weight were −4.3 kg and −0.08 kg in the intervention and control groups, respectively ($P<0.001$). Total and LDL cholesterol fell 13.7 and 13.0 mg/dl in the intervention group and 1.3 and 1.7 mg/dl in the control group ($P<0.001$). HbA1C levels decreased 0.7 percentage point and 0.1 percentage point in the intervention and control group, respectively ($P<0.01$). Conclusions: An 18-week dietary intervention using a low-fat plant-based diet in a corporate setting improves body weight, plasma lipids, and, in individuals with diabetes, glycemic control.